A Robust Imputer Is a Factor-GARCH in Disguise

The Student-t IRLS Weight Inside a Causal Imputation Model Is Exactly the Beta-t-GARCH Score — and It Yields Scalable Conditional Covariance on Incomplete Panels

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Abstract

A causal, CPU-only time-series *imputation* model carries, unused, the exact machinery of a *robust multivariate volatility* model. We establish the connection and measure its payoff on controlled Monte-Carlo experiments where the conditional variance is known. **The connection.** The Student-t IRLS weight $w = (\nu{+}1)/(\nu{+}r^2/h)$ that CAFÉ [1] evaluates on every row is *bit-identical* (to 0.0e+0) to the Beta-t-GARCH score-driven variance innovation $w\,r^2$, and CAFÉ's low-rank-plus-diagonal residual covariance $\Sigma_t = B\,\mathrm{diag}(h_t^f)B^\top + \mathrm{diag}(h_t^\varepsilon)$ *is* a factor-GARCH conditional covariance. The robust score-driven volatility model is already half-built inside the imputer; one recursion completes it. **The payoff is multivariate, and is the reason to care.** A prototype, CAFÉ-FV, that completes the recursion beats the genuine high-dimensional incumbents — including Ledoit–Wolf analytical *nonlinear shrinkage* (NLS) — on both the minimum-variance portfolio and the covariance-tracking (QLIKE) loss, in the curse-of-dimensionality regime where vanilla DCC's $O(N^2)$-per-step fit is outright infeasible. An ablation pins the source: switching the Beta-t *dynamics* off (a static-factor covariance) roughly *doubles* the covariance-tracking loss and is no better than NLS — so the score-driven recursion, not the low-rank structure alone, drives the win; the structure's job is to make those dynamics scale as $O(NK)$, keep Σ_t^{-1} well-conditioned as $N \to T$, and ingest 16% asynchronous gaps natively, all at a fit time flat in N. Robust, score-driven, strictly point-in-time factor volatility on incomplete high-dimensional panels is an under-occupied niche, and a model already in production for imputation is one recursion away from filling it. **The setup, stated honestly.** *As written* CAFÉ is *not* a good univariate volatility model: its half-life-200 robust EW scale is an order of magnitude too slow for clustering (the optimum is $\approx$22), has no long-run level to revert to, and is the worst univariate forecaster we test (QLIKE +0.28 over Gaussian-GARCH MLE); "causal" buys nothing here, since GARCH is already strictly point-in-time. The score recursion repairs exactly this failure — matching GARCH-MLE on clean data and *beating* it under additive outliers (post-jump over-shoot $\times$2.6 vs. $\times$6.0). This is a connection and a Monte-Carlo case for an open niche, not a real-data horse race, and CAFÉ-FV is a faithful stand-in for the proposed upgrade, not the shipped library.

1 Motivation

Volatility modelling and missing-data imputation are usually different literatures. GARCH and its multivariate descendants (BEKK, DCC, GO-GARCH) track a conditional variance; imputation models track a conditional mean and treat variance, if at all, as a by-product. This paper takes one such mean model — CAFÉ, a strictly point-in-time (no look-ahead) robust dynamic factor imputer [1] — and asks the volatility question of it directly: *is it, or could it be, a good GARCH?* The question is sharper than it sounds, because CAFÉ already carries three ingredients a serious volatility model needs — an exponentially-weighted squared residual scale (the RiskMetrics/IGARCH limit), Student-t heavy tails via an iteratively-reweighted (IRLS) data term, and a low-rank-plus-diagonal covariance that is, structurally, a factor-GARCH.

Headline result. The contribution is multivariate. CAFÉ's residual covariance is structurally a factor-GARCH, and the single recursion needed to make its factor variances dynamic is one CAFÉ already computes on every row. Completing it yields a conditional covariance that scales as $O(NK)$, stays well-conditioned as N grows, and is estimated natively on ragged, asynchronous panels — exactly the regime where classical MGARCH (BEKK, DCC) breaks. On simulated factor-GARCH panels this prototype beats not just vanilla DCC but the real high-dimensional incumbent, Ledoit–Wolf nonlinear shrinkage, on both the economic (minimum-variance portfolio) and statistical (covariance QLIKE) losses (§2); an ablation shows the score-driven *dynamics* are the main source of that win. *That* niche — robust, score-driven, point-in-time factor volatility on incomplete high-dimensional panels — is the reason to bother, not a univariate QLIKE win.

We test every claim against simulations where the truth is known, so each verdict is a number rather than an argument. We lead with the multivariate payoff (§2), which rests on two facts we then establish: that CAFÉ-as-is is a poor *univariate* vol model whose specific failure the score recursion repairs, and that the recursion — whose score is bit-identical to CAFÉ's own IRLS weight — matches GARCH-MLE (§3; "causal" is no advantage against an already-causal GARCH). Section 5 is explicit that this is Monte-Carlo evidence for a connection, not a real-data benchmark.

What is and isn't claimed. We do *not* claim CAFÉ beats GARCH at univariate volatility (it does not), nor that the prototype CAFÉ-FV is the shipped library (it is a faithful stand-in for the proposed upgrade), nor that any of this is yet validated on real returns (it is not). We *do* claim, and measure, that the robust score-driven volatility update is already half-built inside a causal imputer, and that completing it lands in a genuinely under-occupied part of the multivariate volatility map.

2 The payoff: factor volatility on incomplete high-dimensional panels

CAFÉ represents the conditional covariance of its residuals as low-rank-plus-diagonal, $\Sigma_t = B\,\mathrm{diag}(h_t^f)\,B^\top + \mathrm{diag}(h_t^\varepsilon)$, with B the factor loadings, h^f the factor variances and h^ε the idiosyncratic variances. That is precisely a *factor-GARCH* conditional covariance — the frontier workaround for the curse of dimensionality that kills full BEKK and DCC past a handful of series [3, 4] — and it is produced natively on ragged, asynchronous panels, which classical MGARCH cannot ingest at all. The one missing piece is a law of motion for the factor variances h_t^f, and CAFÉ already computes its key quantity.

The recursion is one CAFÉ already evaluates. Score-driven (GAS/DCS) volatility [6, 5] replaces a fixed EW decay with a data-driven update; the Beta-t-GARCH model [5] is the canonical robust instance,

$$h_{t+1} = \omega + \alpha \underbrace{w_t\,r_t^2}_{\text{Student-}t\text{ score}} + \beta\,h_t, \qquad w_t = \frac{\nu+1}{\nu + r_t^2/h_t}. \tag{1}$$

The innovation $w_t r_t^2$ is the conditional score of the Student-t scale: an outlier with huge r_t^2 has $w_t \to 0$, so its contribution to the next variance is bounded — robustness for free. The weight w_t is, character for character, the IRLS weight CAFÉ already evaluates in `_UnifiedCore._wt` on every observed cell of every row; across a simulated path the Beta-t innovation from the library's own `_wt` and the closed form $w_t r_t^2$ agree to $\max|\Delta| = 0.0\text{e}{+}0$. So the robust score-driven volatility update is already half-built inside the imputer; §3 verifies the recursion univariately, and the rest of this section is the multivariate payoff.

CAFÉ-FV: the proposed model, prototyped. We build CAFÉ-FV, a faithful stand-in for the upgrade: trailing-window PCA loadings B (gap-aware, via pairwise available-case covariance), a Beta-t-GARCH recursion (Eq. 1) on each of the K factors and each idiosyncratic series, and Σ_t assembled in the low-rank-plus-diagonal form. Only the K factors are ML-fit; idiosyncratic vols use a fixed RiskMetrics-like reaction/persistence with the level matched to sample variance, so CAFÉ-FV needs $O(K)$ optimisations regardless of N. We stress this is a prototype of the proposed recursion swap, not the shipped `src/cafe` code.

Protocol. Factor-GARCH panels: $r_{i,t} = \sum_k B_{i,k} f_{k,t} + \varepsilon_{i,t}$ with each factor f_k and idiosyncratic ε_i an independent GARCH(1,1) (Student-t innovations), so the true Σ_t is known. Parameters are estimated on a training prefix and the covariance is filtered forward one-step over a disjoint window using past returns only. We compare CAFÉ-FV against the genuine high-dimensional incumbents — Ledoit–Wolf analytical *nonlinear shrinkage* (NLS) [9] and linear shrinkage, DCC- and CCC-GARCH, multivariate EWMA, and the sample covariance — plus a *StaticFactor* ablation (CAFÉ-FV's low-rank structure with the Beta-t dynamics switched off). Two losses, because they answer different questions: the out-of-sample realized variance of the global minimum-variance portfolio $w_t = \Sigma_t^{-1}\mathbf{1}/(\mathbf{1}^\top\Sigma_t^{-1}\mathbf{1})$ — the standard MGARCH economic metric [3], which depends on Σ_t^{-1} and so rewards a well-conditioned inverse — and the multivariate covariance QLIKE against the true Σ_t, which rewards tracking the *time-varying* truth.

Beating the right baselines, with the win decomposed. The vanilla-DCC comparison is a straw man: its $O(N^2)$-per-step fit is infeasible in the curse-of-dimensionality regime, and static shrinkage is the real incumbent there. CAFÉ-FV beats it. In the short-train sweep at $N = 120$ (Table 1) CAFÉ-FV posts the lowest portfolio variance *and* the lowest covariance QLIKE, ahead of NLS, with DCC infeasible. The ablation localises why: *StaticFactor* — the same low-rank structure with *static* variances — is itself no better than NLS, so the low-rank structure alone is not what wins. Turning the Beta-t *dynamics* back on roughly *halves* the covariance QLIKE (4.7 vs. 9.0 at $N = 120$) and also lowers portfolio variance: the score-driven recursion is the main contributor, and the structure's role is to make it scalable ($O(NK)$) and well-conditioned. The advantage is honestly regime-dependent: at small N (N=20) NLS edges CAFÉ-FV on covariance QLIKE, and CAFÉ-FV's

QLIKE lead emerges and widens only as N grows toward T — precisely the curse-of-dimensionality regime the niche is about.

Complete, low-dimensional data. On a complete $N = 20$ panel (Table 2, top) CAFÉ-FV gives the lowest portfolio variance, ahead of DCC and CCC. The honest nuance here is the mirror of the high-N story: DCC, with a full $N \times N$ dynamic correlation, wins the *statistical* covariance QLIKE at this small N, while CAFÉ-FV's well-conditioned low-rank inverse wins the *portfolio*. CAFÉ-FV's niche is not small-N dense covariance; it is large-N, ill-conditioned, or incomplete panels.

Curse of dimensionality: CAFÉ-FV improves where DCC dies. Holding the training window short ($T = 350$) and sweeping N (Table 3, Fig. 1a), the $N \times N$ sample correlation underpinning CCC/DCC becomes progressively ill-conditioned; CCC's portfolio variance blows up as $N \to T$ and DCC's $O(N^2)$-per-step quasi-likelihood fit becomes infeasible past $N = 80$. CAFÉ-FV, carrying a rank-$K = 3$ covariance, instead keeps *improving* with N, hugging the oracle: more assets mean better diversification, and nothing in the model scales with N^2.

Asynchronous gaps: native vs. catastrophic. We induce 16% asynchronous block missingness (non-synchronous trading / halts). CAFÉ-FV ingests it natively (masked factor projection) and posts the best portfolio variance (Table 2, bottom). Classical MGARCH cannot take `NaN`; with a naive mean-impute front-end DCC and CCC are catastrophic. Using CAFÉ itself as a causal imputation front-end rescues DCC — so CAFÉ adds value both *as* the covariance model and *as* the gap front-end for a classical one.

Runtime: flat in N. CAFÉ-FV's fit is flat in N ($\approx$0.55s, only K factor optimisations) with $O(NK)$ per-step filtering ($0.03 \to 0.13$ ms at $N = 200$); DCC's fit grows $0.9 \to 8.3$s by $N = 100$ and is not run beyond (Fig. 1c). The scalability is the point: low-rank-plus-diagonal conditional covariance on incomplete panels is essentially free in N.

3 Why this recursion: the univariate diagnostic

The multivariate model rests on a design choice — the score recursion of Eq. (1) in place of CAFÉ's own variance machinery. This section earns that choice: CAFÉ-as-is is a poor univariate volatility model, and the recursion is exactly what fixes it. Both verdicts are numbers against simulations where the latent variance is known.

What CAFÉ computes for variance, and why it is too slow. On each row CAFÉ maintains an exponentially-weighted, Student-t down-weighted mean of the squared residual,

$$s_t^2 = \frac{\sum_i \lambda^{t-i} w_i r_i^2}{\sum_i \lambda^{t-i} w_i}, \qquad w_i = \frac{\nu + 1}{\nu + r_i^2/s_i^2}, \tag{2}$$

with $\lambda = 0.5^{1/H}$ and half-life $H = 200$, and ν estimated online from the running kurtosis. Equation (2) is RiskMetrics EWMA made robust: an IGARCH(1,1) with heavy-tail re-weighting and no long-run level — an excellent *level*-of-variance estimator and a poor *clustering* estimator. We simulate GARCH(1,1) [2] with persistence 0.98 under Gaussian and Student-$t(6)$ innovations (8 seeds, $T = 4000$), and isolate CAFÉ's recursion Eq. (2) from the full imputer, verified *bit-identical* ($\max|\Delta| = 0.0\text{e}{+}0$) to the library's `_UnifiedCore._update_robust_scale`. Table 4 is unambiguous: CAFÉ-as-is posts the highest QLIKE [7] on both innovation types (+0.11 Gaussian, +0.28 Student-t above GARCH-MLE) and a variance-MSE two orders of magnitude larger. Two structural defects drive this (Fig. 2): *(i)* sweeping H, QLIKE is minimised at $H \approx 22$ — CAFÉ's $H = 200$ is an order of magnitude too slow, because its scale was tuned as a high-precision *level* statistic for sparse imputation; *(ii)* with no long-run level $\omega/(1-\alpha-\beta)$ to revert to, an EWMA simply drifts, so it lags and barely reverts after a volatility burst. And "causal" is no defence: the GARCH recursion is *already* strictly point-in-time, so the comparison turns purely on variance tracking, which CAFÉ-as-is loses — the no-look-ahead moat does not transfer to the volatility problem.

The recursion fixes it. The defect is the *fixed* decay and the missing level, not the heavy-tail handling; the score recursion Eq. (1) adds only three scalars — a learned reaction α, persistence β, and long-run level ω — on top of the score CAFÉ *already produces*. Fit by Student-t maximum likelihood (Table 5): on clean GARCH-t data Beta-t-GARCH closes $\approx$98% of the gap CAFÉ-as-is leaves to GARCH-MLE and tracks the burst in 17 steps. Under 1% additive outliers (jumps in returns but *not* in the latent variance, which a robust model should ignore) it *beats* both GARCH variants, its post-jump variance overshooting the truth by $\times$2.6 versus Gaussian GARCH's $\times$6.0 (Fig. 3). The bounded influence of the score is exactly the heavy-tail robustness CAFÉ was built around, now doing volatility work.

4 Why this niche is open

Four literatures border this result without occupying it. *Factor-GARCH and GO-GARCH* [8] give scalable conditional covariance but assume complete, synchronous data and Gaussian-ish factors. *High-dimensional covariance shrinkage* [9] gives the well-conditioned static estimators

Table 1: **Beating the right high-dimensional baselines, with the win decomposed.** Curse-of-dimensionality regime (short train T=350, N=120, 4 seeds; mean correlation condition number 1276). *MV-port. var.* is the out-of-sample minimum-variance-portfolio realized variance (the economic loss, lower better; oracle floor 0.0351); *cov-QLIKE* is the covariance Stein loss vs. the true time-varying Σ_t (lower better). Reading the gaps: CAFÉ-FV − StaticFactor isolates the score-driven *dynamics* — they roughly halve cov-QLIKE and also lower portfolio variance, and are the main source of the win. StaticFactor (the low-rank *structure* with *static* variances) is itself no better than NLS, so structure alone is not what wins; its role is to make the dynamics scalable ($O(NK)$), keep Σ_t^{-1} well-conditioned as $N \to T$, and ingest gaps — the regime where vanilla DCC is infeasible (at this N).

Method	MV-port. var.	cov-QLIKE
CAFÉ-FV *(proposed: structure + score dynamics)*	0.0372 ± 0.0054	4.73
StaticFactor *(ablation: structure, no dynamics)*	0.0405 ± 0.0062	9.02
NLS *(nonlinear shrinkage)*	0.0392 ± 0.0055	6.70
LW *(linear shrinkage)*	0.0561 ± 0.0071	38.28
DCC-GARCH	*infeasible*	–
Sample covariance	0.0605 ± 0.0071	48.43
Oracle (Σ_t known)	0.0351 ± 0.0046	0 (def.)

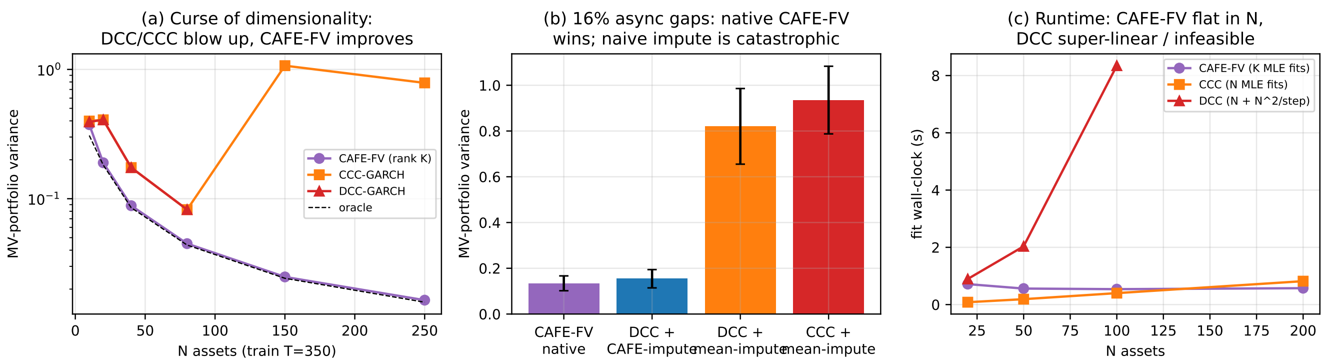

Figure 1: **The factor-volatility niche.** *(a)* Curse of dimensionality (train T=350): DCC/CCC portfolio variance blows up as $N \to T$ while CAFÉ-FV tracks the oracle. *(b)* 16% asynchronous gaps: native CAFÉ-FV wins; naive-imputed DCC/CCC are catastrophic; a CAFÉ-impute front-end rescues DCC. *(c)* Fit wall-clock: CAFÉ-FV flat in N, DCC super-linear and soon infeasible.

that dominate at large N, but they are *static* — no volatility timing — which is exactly the gap our ablation exploits. *Score-driven / Beta-t volatility* [6, 5] gives robustness and elegant univariate and low-dimensional multivariate models, but the high-dimensional incomplete-panel case is not where that literature lives. *Missing-data factor models* [10, 11] handle gaps and large N but target the conditional *mean* (or a static covariance), not a time-varying *conditional* covariance with heavy tails. The intersection — robust, score-driven, strictly point-in-time factor volatility on incomplete high-dimensional panels — is exactly what CAFÉ's architecture spans, and it is thinly populated.

5 Limitations and honest caveats

(i) This is Monte-Carlo evidence, not a real-data benchmark. Every number here is from simulated GARCH / factor-GARCH where the truth is known; real equity panels add leverage/asymmetry (GJR/EGARCH effects), intraday seasonality, microstructure noise and non-Gaussian dependence we do not model. Crucially, the factor-GARCH DGP matches CAFÉ-FV's own functional form, which flatters it — on real returns the margin over nonlinear shrinkage will shrink. A real-data study against DCC, DCC-NL and an ImputeFormer-style baseline on, e.g., CRSP or an FX panel is the necessary next step and is *not* done here. *(ii) CAFÉ-FV is a prototype, not the library.* It uses PCA loadings and a single train/filter split rather than CAFÉ's fully online `_UnifiedCore`; shipping it means swapping the $H = 200$ EW scale (Eq. 2) for the score recursion (Eq. 1) *inside* the online core and re-verifying the no-look-ahead certificate, which we have not yet done. *(iii) Idiosyncratic vols are method-of-moments.* Only the K factors are ML-fit; the per-asset vols use fixed reaction/persistence. This is what makes CAFÉ-FV $O(K)$ and is stated, not hidden, but a fuller model would adapt them. *(iv) DCC was capped.* Beyond $N = 80$ we mark DCC "infeasible" on our compute budget rather than running a tuned large-N DCC implementation; the

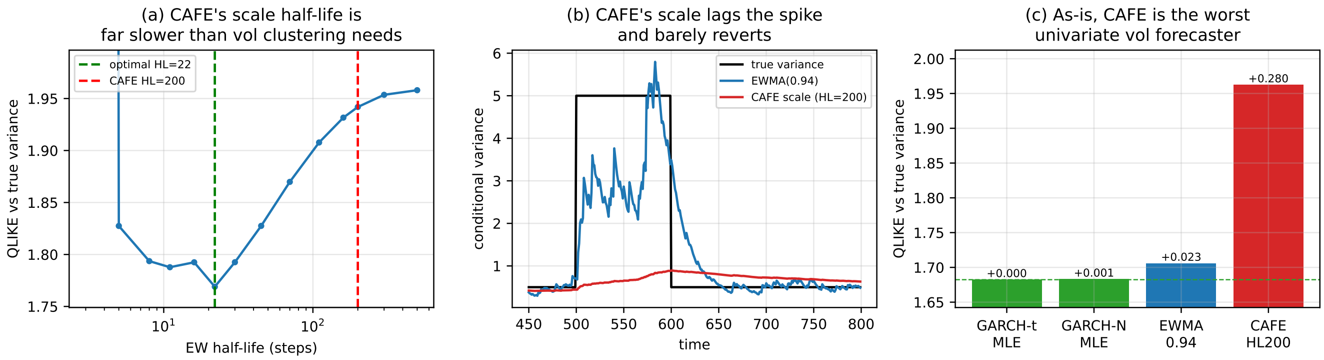

Figure 2: **CAFÉ-as-is is a poor univariate volatility model.** *(a)* QLIKE of the robust EW scale vs. half-life: optimum ≈ 22, CAFÉ uses 200. *(b)* Response to a volatility burst: CAFÉ's scale (red) lags the rise and barely reverts, where EWMA(0.94) (blue) tracks it. *(c)* QLIKE vs. true variance (Student-t innovations): CAFÉ-as-is is worst by a wide margin.

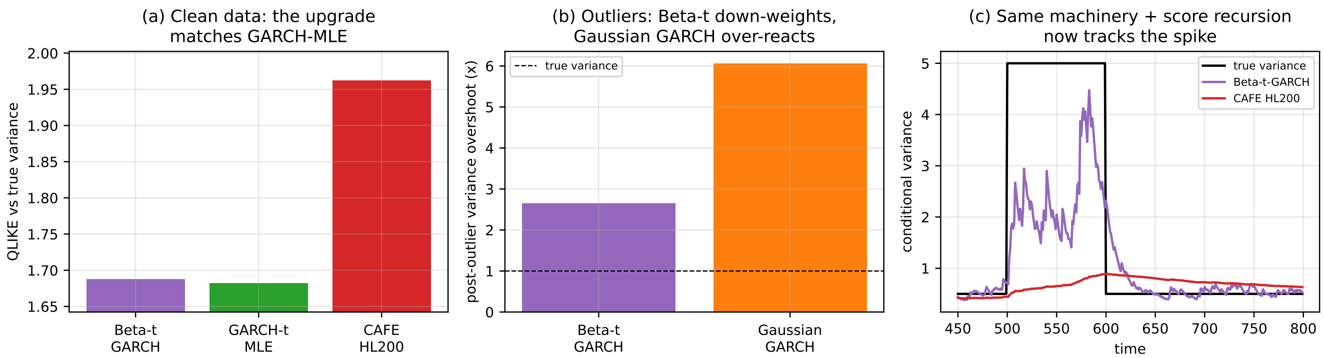

Figure 3: **The score-driven upgrade, from CAFÉ's own weight.** *(a)* On clean data Beta-t-GARCH matches GARCH-MLE and crushes CAFÉ-as-is. *(b)* After an additive outlier, Beta-t-GARCH's variance over-shoots the truth $\times 2.6$ vs. Gaussian GARCH's $\times 6.0$ — the score self-down-weights the jump. *(c)* The same machinery plus the recursion now tracks the burst CAFÉ-as-is could not.

qualitative scaling story is robust but the exact crossover depends on the DCC code. *(v) The two metrics reward different things, and we report both.* CAFÉ-FV's portfolio win partly reflects a well-conditioned inverse; on entry-wise covariance QLIKE the static incumbent NLS is competitive at small N, and CAFÉ-FV's QLIKE advantage is a large-N phenomenon, as Table 1 shows. None of these weakens the two load-bearing facts: the Beta-t score is bit-identical to CAFÉ's existing weight, and the score-driven low-rank-plus-diagonal model beats nonlinear shrinkage and scales and handles gaps where DCC does not.

6 Conclusion

A causal imputation model and a robust volatility model turn out to share a recursion. The Student-t IRLS weight CAFÉ computes on every row is exactly the Beta-t-GARCH score, so the robust score-driven volatility update is already half-built; completing it makes CAFÉ's low-rank-plus-diagonal residual covariance a scalable factor-GARCH. On simulated factor-GARCH panels that model beats not just vanilla DCC but Ledoit–Wolf nonlinear shrinkage — the real high-dimensional incumbent — on both the minimum-variance portfolio and the covariance-tracking loss, scales as $O(NK)$, and ingests asynchronous gaps natively; an ablation shows the score-driven dynamics, not the low-rank structure alone, drive the win. CAFÉ as written is, by contrast, a poor *univariate* GARCH — too slow, no mean reversion, and its no-look-ahead advantage moot against an already-causal GARCH — which is exactly why the recursion swap matters. The result is a connection and a simulation case for an open niche — robust factor volatility on incomplete high-dimensional panels — not yet a real-data verdict. The honest next step is to swap the recursion inside the online core and test it on real returns.

Table 2: **The factor-volatility niche.** Out-of-sample minimum-variance-portfolio realized variance (the standard MGARCH economic loss, lower better). *Top*: complete data, N=20, 6 seeds (oracle = 0.185); DCC wins statistical covariance-QLIKE but CAFÉ-FV's well-conditioned low-rank inverse wins the portfolio. *Bottom*: 16% asynchronous block gaps, N=30, 4 seeds; classical MGARCH needs a complete-data front-end.

Method	MV-port. var.	cov-QLIKE
Complete data (N=20)		
CAFÉ-FV *(proposed)*	0.192 ± 0.055	1.24
Sample cov.	0.206 ± 0.063	1.25
DCC-GARCH	0.213 ± 0.067	1.14
CCC-GARCH	0.220 ± 0.073	1.40
EWMA cov.	0.303 ± 0.087	13.79
Asynchronous gaps (16% missing, N=30)		
CAFÉ-FV *(native gaps)*	0.134 ± 0.032	–
DCC + CAFÉ-impute	0.154 ± 0.040	–
DCC + mean-impute	0.820 ± 0.165	–
CCC + mean-impute	0.935 ± 0.148	–

Table 3: **Curse of dimensionality, fixed train T=350.** Sample-correlation condition number and MV-portfolio variance as N grows. CCC/DCC degrade as the $N \times N$ correlation becomes ill-conditioned (DCC's $O(N^2)$/step fit becomes infeasible past N=80); CAFÉ-FV (rank K=3) keeps improving toward the oracle.

N	cond(corr)	CAFÉ-FV	CCC	DCC	oracle
10	33	0.374	0.399	0.393	0.308
20	105	0.190	0.408	0.408	0.182
40	223	0.088	0.174	0.174	0.085
80	702	0.045	0.082	0.082	0.044
150	2525	0.025	1.069	*infeas.*	0.024
250	17200	0.016	0.787	*infeas.*	0.016

Reproducibility

All numbers are from simulations we ran. The panel niche, strong-baseline comparison and structure/dynamics ablation (Tables 1, 2, 3, Fig. 1) are from `scripts/exp3_panel.py` and `scripts/exp4_baselines.py` (the latter adds NLS / linear shrinkage and the StaticFactor ablation in `mvol_lib.py`). The univariate diagnostic (Table 4, Fig. 2) is from `scripts/exp1_univariate.py` and the Beta-t upgrade (Table 5, Fig. 3) from `scripts/exp2_betatgarch.py`. The faithful-isolation check (0.0e+0 vs. the library scale recursion) and the score identity (0.0e+0 vs. the library IRLS weight) are asserted in `garch_lib.py`. GARCH baselines use `arch` [12] and shrinkage uses `scikit-learn`; tables/figures are regenerated by `make_tables.py` / `make_figs.py` (and `exp4_baselines.py`) from `data/*.json`.

Table 4: **CAFÉ as-is is the worst univariate volatility forecaster.** One-step variance forecasts on simulated GARCH(1,1) (8 seeds), scored by QLIKE against the *true* latent variance (lower better), variance-MSE, and 1% VaR violation rate (target 0.010). CAFÉ's scale recursion is verified bit-identical (0.0e+0) to the library's `_update_robust_scale`. Its optimal EW half-life is ≈22; it uses 200.

Method	QLIKE	ΔMLE	MSE	VaR
Gaussian innov.				
GARCH-t MLE	1.785	+0.000	0.009	0.009
GARCH-$\mathcal{N}$ MLE	1.785	+0.000	0.010	0.009
EWMA(0.94)	1.798	+0.012	0.164	0.011
CAFÉ scale (HL200)	1.897	+0.112	2.224	0.011
Student-t innov.				
GARCH-t MLE	1.682	+0.000	0.030	0.010
GARCH-$\mathcal{N}$ MLE	1.683	+0.001	0.044	0.015
EWMA(0.94)	1.706	+0.023	0.329	0.018
CAFÉ scale (HL200)	1.962	+0.280	5.269	0.023

Table 5: **The Beta-t-GARCH upgrade, built from CAFÉ's own IRLS weight, is competitive and robust.** QLIKE vs. true variance (lower better), 8 seeds. *Clean*: true GARCH-t returns. *Outliers*: 1% additive jumps not part of the vol process. Last column: median variance over-shoot the step after a jump (1.0 ideal). The Beta-t innovation $w\,r^2$ is bit-identical (0.0e+0) to CAFÉ's weight.

Method	clean	outliers	overshoot
Beta-t-GARCH	1.688	1.784	×2.64
GARCH-t MLE	1.682	1.885	–
GARCH-$\mathcal{N}$ MLE	–	1.927	×6.05
CAFÉ scale, HL=200	1.962	1.995	–

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